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SUPERSEDING
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Fed. Spec. UU-P-525B
July 29, 1963

FEDERAL SPECIFICATION

PAPER SET, MANIFOLD AND CARBON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers paper sets, each set consisting of one sheet of manifold paper overlaid with a sheet of one-time carbon paper, for use with standard commercial typewriters of the manual and electrical types.

1.2 Classification.

1.2.1 Type and grade. The paper set shall be of one type and grade.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-R-350 - Ribbon, Typewriter and Composing Machine, Changeable Type-Plate Style, Polyethylene.
NN-P-71 - Pallets, Material Handling, Wood, Double Faced, Stringer Construction.
UU-P-121 - Paper, Bond and Writing, White and Colored.
UU-P-328 - Paper, Manifold.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Technical Association of the Pulp and Paper Industry (TAPPI) Standards:

T 402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets and Related Products.
T 410 - Weight Per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 414 - Internal Tearing Resistance of Paper.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Manifold paper. Manifold paper shall conform to the requirements for type V of UU-P-328. The color of the manifold paper shall be white, blue, green, pink, salmon, or yellow, as specified (see 4.3.1.2, 6.2, and 6.3).

3.1.2 Carbon paper. The carbon paper shall be one-time carbon and shall consist of carbonizing paper sufficiently coated with suitable carbon ink and other materials necessary to provide the copying qualities specified in 3.3 and to make a permanent record.

3.1.2.1 Physical properties. The basis weight and tearing resistance of the carbonizing paper shall be in accordance with the following:

Weight in pounds, decoated 20 x 30 - 500 sheets.....8 pounds.
Tearing resistance: Average, each direction not less than 5 grams.

A tolerance of minus 5 percent or plus 6 percent in weight will be permitted. Test may be made without removing the carbon coating.

3.1.2.2 Coating. The coating shall be smooth and uniform and shall cover the entire surface of the carbon paper except for the uncoated ends of the length dimension as indicated in 3.2.1. The coating shall not separate or flake when tested as specified in 4.3.2. The length of the coated portion of the carbon paper shall be as follows:

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Manifold sheet size	Coating length 1/
8 by 7 inches	6-1/4 inches
8 by 10-1/2 inches	9-3/4 inches
8 by 13 inches	12-1/4 inches
8-1/2 by 14 inches	13-1/4 inches
8-1/2 by 11 inches	10-1/4 inches

1/ A tolerance of ± 1/16 inch shall be permitted.

3.1.2.3 Finish. The carbon paper shall be furnished in a finish which shall meet the performance requirements specified in 3.3.

3.1.2.4 Color. The color of reproductions produced by the carbon paper shall be black.

3.1.2.5 Smudging. There shall be no smudge marks on the manifold copies after the paper sets have been subjected to the performance test specified in 4.3.2.2.

3.2 Construction. The paper set shall consist of one sheet of manifold paper (see 3.1.1) and one sheet of carbon paper (see 3.1.2), assembled as specified hereinafter. Sets attached to each other or in pad form are not acceptable.

3.2.1 Assembly. Each set shall be assembled with the coated side of the carbon sheet facing the manifold paper. The carbon sheet shall be attached to the manifold sheet at the top of the sheet, by such method that it will remain attached during normal handling. Also, the manner of attachment shall be such as to permit the extraction of carbon sheets (or manifold sheets) assembled and typed at one time as specified in 4.3.2.2, easily and without tearing of either the carbon or manifold copies. Adhesive, when used, shall be affixed to the detachable and unusable portion of the set, and there shall be no excessive adhesive marks or stains on the manifold paper after the paper sets have been tested as described in 4.3.2.2.

3.2.1.1 Size. Paper sets shall be furnished in the following sizes (based upon the dimensions of the manifold paper), as specified (see 6.2): 8 by 7 inches, 8 by 10-1/2 inches, 8 by 13 inches, and 8-1/2 by 14 inches, 8-1/2 by 11 inches. Dimensional tolerances for the manifold paper shall be as specified in UU-P-328.

3.2.1.2 Bottom of set. The carbon and manifold sheets shall be set off, leaving an uncoated or clean edge of carbon paper extending 1/2 inch, ± 1/16 inch, beyond the bottom edge of the manifold sheet.

3.2.1.3 Top of set. The distance from the top edge of the manifold sheet to the top edge of the coated portion of the carbon paper shall be 11/16 inch, ± 1/16 inch. The distance from the top edge of the carbon paper to the top edge of the coated portion (uncoated edge) shall be 3/8 inch, ± 1/16 inch.

3.2.1.4 Alignment. The alignment of the carbon paper sheet with the side edges of the manifold sheet shall be within 1/32 inch.

3.2.2 Bid samples. When specified (see 6.2), bid samples shall be submitted for the sizes and color in the quantities specified, for examination and testing as specified in 6.4.

3.3 Manifolding performance. The carbon paper shall provide at one time not less than eight manifold copies. The intensity and sharpness of the images produced on all manifold copies shall be equal to or better than the Government exhibits for manifolding performance, when tested in accordance with 4.3.2 (see 6.5).

3.4 Workmanship. The workmanship shall be in accordance with standard commercial practice. The carbon paper shall be free from uncoated areas, hard spots, wrinkles, welts, lumps, and any other defects or irregularities which affect its appearance and which might affect its serviceability.

3.5 Quality. The quality shall conform to levels established in this specification. The occurrence of defects shall not exceed the number permitted under the acceptable quality levels (AQL's) specified in section 4 herein for all examinations and tests of materials, components, and end items as applicable.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.1.2 Inspection of components and materials. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with this specification or referenced subsidiary specifications and standards.

4.2 Inspection of the end item.

4.2.1 Sampling. Samples of shipping containers, packages (unit boxes of paper sets), and paper sets shall be examined for the defects set forth in the applicable subparagraphs. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. For sampling purposes, the lot shall consist of all paper sets of the same size and color presented for inspection at one time.

TABLE I. Sampling procedure

Examination paragraph	Lot size units	Inspection level	AQL defects per hundred units
4.2.1.1	Shipping containers	S-2	2.5
4.2.1.1.1	Packages 1/	S-3	4.0
4.2.1.2	Paper sets 1/	S-3	2.5
4.2.1.3	Paper sets 1/	S-3	2.5

1/ Packages sampled shall be from the shipping containers selected for examination in 4.2.1.1.

4.2.1.1 Examination of preparation for delivery. An examination shall be made to determine that markings, materials, and workmanship comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, closed and sealed. Defects shall be scored as listed below.

Examine	Defect
Marking (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Material	Any nonconforming component; component missing, damaged or otherwise defective.
Containers	Not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps; loose or inadequate sealing, strapping, or stapling.
Contents	Number of packages (unit boxes) per shipping container is less than the specified. Not packed as specified.
Pallet	Not materials or contruction as specified; gross weight exceeds 3000 pounds.

4.2.1.1.1 Package (unit box). The sample unit for this examination shall be one package of paper sets. Approximately an equal number of packages shall be selected from the shipping containers selected under 4.2.1.1. Defects shall be scored as listed below.

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Examine	Defect
Material and workmanship	Not as specified; damaged or otherwise defective; not new.
Marking (interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Count	Less than 500 sheets per package, as specified.

4.2.1.2 Examination of the end item for dimensional defects. The sample unit shall be one paper set (manifold and carbon), and not more than two sets shall be selected from any package. Defects shall be scored as listed below.

Examine	Defect
Manifold paper	Not as specified; undersize; oversize by more than 1/16 inch along any edge from size specified.
Carbon per Length	Length of coated portion not within size and tolerance for applicable manifold sheet size.
Alignment	Not within 1/32 inch of side edges of manifold sheet.
Assembly	Uncoated bottom edge extends more than 9/16 inch or less than 7/16 inch beyond bottom edge of manifold sheet.
	Top edge of coated portion more than 3/4 inch, or less than 5/8 inch for top edge of manifold sheet.
	Uncoated top edge more than 7/16 inch or less than 5/16 inch.

4.2.1.3 Examination of the end item for defects in construction and appearance. The sample unit for this examination shall be one paper set and not more than two sample units shall be selected from any packages. Defects shall be scored as listed below.

Examine	Defect
Manifold paper	Presence of tears, wrinkles, folds, flakes, cuts, lumps, dirt, grit, or foreign matter.
	Partial sheet.
	Coating not black.
Color	Not color specified.
Construction	Coated side of carbon paper not facing manifold sheet.
	Carbon paper not attached to manifold paper.
	Excessive adhesive; adhesion marks or stains on manifold paper.
Appearance	Not smoothly or uniformly coated; area of insufficient or no coating, except as specified.

4.3 Testing of the end item. Paper sets shall be tested for performance and the physical properties of the carbon and manifold paper specified in section 3, for each lot submitted for inspection at one time. The sample unit for these tests shall consist of 28 paper sets selected in consecutive order from one package. Five sample units randomly selected from the shipping containers selected under 4.2.1.1 shall be tested. For each sample unit, eight paper sets shall be used for the test specified in 4.3.2 and the remainder reserved for the tests specified in 4.3.1. There shall be no failure to meet the requirements specified in section 3 for each sample unit tested.

4.3.1 Physical properties.

4.3.1.1 Carbon paper.

4.3.1.1.1 Weight (carbonizing paper), decoated. For the determination of the basis weight of the decoated carbonizing paper, the carbon coating shall be removed with benzol or other suitable solvent, aided by gentle scrubbing. After the coating has been removed the carbonizing paper shall be dried and conditioned according to atmospheric conditions described in TAPPI T402. Basis weight shall then be determined as described in TAPPI T410.

4.3.1.1.2 Tearing resistance. After conditioning in an atmosphere in accordance with TAPPI T402, tearing resistance of the carbon paper shall be determined in accordance with TAPPI T414.

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4.3.1.2 Manifold paper. The physical properties of the manifold paper shall conform to the requirements set forth in UU-P-328, when tested as specified therein. In lieu of using his own facilities, the supplier may submit a report of analysis of the manifold paper from a laboratory acceptable to the Government whether an independent test facility or the laboratory of the raw material supplier, as evidence of compliance with the specification requirements.

4.3.2 Performance (paper sets).

4.3.2.1 Apparatus. The paper sets shall be tested in an electric typewriter with pica type. The typewriter shall be equipped with a typewriter ribbon conforming to L-R-350 and a platen having a hardness reading of between 89 to 93, when measured with a Shore durometer, type A. The type shall be thoroughly cleaned before each test. The typewriter key force shall be adjusted so that lower case letters shall barely emboss one sheet of 32-pound bond paper (17 x 22 - 1000), conforming to type III of UU-P-121, when the typewriter key impression indicator is set at the middle of its range.

4.3.2.2 Procedure. A first sheet of 40-pound bond paper (17 x 22 - 1000), conforming to the requirements of type III of UU-P-121 and eight sample sets of manifold and carbon paper shall be assembled in the usual way. Examine the sets for separation of carbon and manifold paper. The typewriter impression indicator setting and the platen setting shall be set as recommended by the machine manufacturer for eight copies. The sets shall be inserted in the typewriter and the entire keyboard, both upper and lower cases, shall be written twice over in unrelated order, followed by complete sentences to provide coverage of not less than 75 percent of the carbon area. The bond first sheet and manifold sets shall be removed from the machine and the eight carbons (or manifold sheets) shall be extracted at one time. Examine for ease of carbon paper removal (without tearing of either carbon or manifold copies), coating flaking or smudging, and manifolding performance.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Five hundred paper sets of one size and color shall be packaged in snug-fitting box conforming to style III or VIII, type G, class i of PPP-B-566; type I, style 3 or 4 of PPP-B-676 or style A of PPP-B-665. Closure shall be effected with a 2 inch minimum width gummed paper tape conforming to type III, class B of PPP-T-45 applied over the full perimeter seam of each box.

5.1.2 Level B. Five hundred paper sets of one size and color shall be packaged in a full-telescope box. Box base and cover shall be constructed from stock of 24 point full-bending chipboard or better. Base box and cover box shall be of the end-wall-overlap style, with overlap secured to end wall flaps by adhesive suitable for the purpose. Closure shall be as specified in 5.1.1.

5.1.2.1 Alternate. Five hundred paper sets of one size and color shall be packaged in a snug fitting folding paperboard box conforming with PPP-B-566, style 1, type A, class A. The caliper of the material used in making the box shall be not less than .024 inch. The box shall be constructed with a device to permit easy opening and which will protect the end of the paper after the box has been opened.

5.1.3 Level C. Paper sets shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Paper sets packaged as specified in 5.1 shall be packed in accordance with level A, B, or C as specified (see 6.2).

5.2.1 Level A. Ten unit boxes packaged as specified in 5.1.1 and of one size and color shall be packed in a fiberboard box conforming to class weather resistant, V3c or V3s, style RSC of PPP-B-636. End loading (boxes standing on end) shall not be acceptable. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat sheet or non-metallic strapping in accordance with the appendix of PPP-B-636.

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5.2.2 Level B. Ten unit boxes packaged as specified in 5.1 and of one size and color shall be packed in a fiberboard box conforming to PPP-B-636, class domestic, style RSC, constructed of fiberboard of not less than 200 pounds per square inch (psi). End loading (boxes standing on end) shall not be acceptable. Closure shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Paper sets packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Palletization. When specified (see 6.2) the boxes of paper sets shall be palletized. The boxes shall be assembled into a load which shall not exceed 3,000 pounds. The load shall not be more than 1-1/2 inch longer in either length or width than the pallet. Strapping of the load is recommended.

The pallet shall be a non-returnable, non-reversible pallet whose smallest dimension shall be not less than 32 inches and the largest shall not exceed 54 inches.

The lumber shall be one of the species listed in NN-P-71, group 2 or group 3. The deck boards shall be not less than 5/8 by 3 inches actual size. There shall be not more than 3 inches of space between the top deck boards.

When the pallet is of the block type, there shall be nine hardwood blocks 3-1/2 x 3-1/2 x 3-1/2 inches in size. Boards, minimum size 5/8 by 3 inches actual size, shall be placed on the top and bottom of the hardwood blocks at right angles to the deck boards. In addition, three boards, minimum size 5/8 by 3 inches actual size, shall be placed on the bottom of the pallet, running parallel to the deck boards. These boards may be placed either above or below the boards on the bottom of the pallet which runs at right angles to the deck boards and shall be located under the blocks.

When the pallet is of the stringer type there shall be three stringers of 2 x 4 inch lumber. The lumber shall conform to NN-P-71, group 2 or 3. A stringer shall be placed within three inches of the end of each of the deck boards and the other spaced midway between two outer stringers. The stringers shall be cut out a minimum of 1-3/8 x 9 inches starting approximately 6 inches from each end of the stringers to provide for four-way entry with fork lifts. A minimum of three deck boards, evenly spaced, shall be placed on the bottom of the stringers.

Assembly of the pallet shall be by nailing, using cement coated, chemically etched or mechanically deformed (screw type or annular-ring) nails.

5.4 Marking. In addition to any special marking required by the procurement document, marking of interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.2).

6. NOTES

6.1 Intended use. Paper sets are to be used for carbon copies of typed material.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of set required (see 3.2.1.1).
(c) Color of manifold paper (see 3.1.1).
(d) Bid samples, if required and quantity required (see 3.2.2).
(e) Level of packaging and packing required (see 5.1 and 5.2).
(f) Palletization if required (see 5.3).
(g) Marking required (see 5.4).

6.3 Comparison sample. A standard sample required as specified in UU-P-328, may be obtained from the procuring agency issuing the invitation for bids (see 3.1.1).

6.4 Bid samples. When bid samples are required, they shall be examined for conformance to 3.2.1, 3.3, and 3.4.

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6.5 Exhibits of manifolding performance specimens. Specimen exhibits of carbon paper reproductions of acceptable intensity and sharpness are available for examination for comparison purposes at the Quality Control Division Offices, in all General Services Administration Regional Offices. These specimens reflect the standard of quality required by the Government, in accordance with the requirements of the specification.

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